

TIMELESS TECHNOLOGY SOLUTIONS

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PROPOSAL IN RESPONSE TO

Inventory & Warehouse Management System

Submitted To:	Meridian Distributors, LLC
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1. EXECUTIVE SUMMARY

Timeless Technology Solutions is pleased to respond to Meridian Distributors RFP No. MDL-2023-047 for a custom Inventory and Warehouse Management System (IWMS). Meridian operates three distribution centers across Kentucky and Tennessee, processing over 8,000 SKUs and 350 outbound shipments daily. The current system is a legacy desktop application from 2009 that lacks barcode scanning, real-time stock visibility, and integration with Meridian's QuickBooks accounting software.

TTS proposes a cloud-based IWMS built on a **Python/FastAPI backend** with a **React dashboard**, native barcode-scanner support via a companion Android/iOS app, and bi-directional QuickBooks Online integration. Total project cost is **\$142,000** delivered over **22 weeks**.

2. UNDERSTANDING OF REQUIREMENTS

Based on the RFP and our discovery call on September 28, 2023, TTS understands the primary requirements to be:

- Real-time inventory tracking across all three warehouse locations with sub-location (aisle/bay/shelf) granularity.
 - Barcode and QR-code scanning for inbound receipts, put-away, picking, and outbound shipping confirmation.
 - Automated reorder-point alerts with configurable lead-time and safety-stock rules per SKU and supplier.
 - Integration with QuickBooks Online for purchase order sync, cost-of-goods accounting, and accounts-payable matching.
 - Carrier label printing (UPS, FedEx, LTL) and shipment tracking lookup.
 - Role-based access: Warehouse Associate, Supervisor, Purchasing, Accounting, Admin.
 - Reporting: daily stock valuation, slow-mover analysis, shrinkage/adjustment log, fulfillment accuracy rate.
 - Data migration from the legacy system (estimated 8,000 active SKUs, 5 years of transaction history).
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3. TECHNICAL APPROACH

System Architecture

- **Backend:** Python 3.12 + FastAPI; SQLAlchemy ORM; Alembic migrations; Celery + Redis for async tasks (label generation, reorder alerts, QuickBooks sync).
- **Database:** PostgreSQL 16 with partitioned transaction tables for performance at scale; full-text search via pg_tsvector for SKU/product lookup.
- **Web Dashboard:** React 18 + Vite; TanStack Table for grid-heavy views; Recharts for inventory analytics; print-optimized CSS for pick lists and manifests.
- **Mobile Scanner App:** React Native (iOS + Android); camera-based barcode scanner via Expo; offline queue with background sync for warehouse dead zones.
- **Integrations:** QuickBooks Online REST API (OAuth 2.0); UPS/FedEx REST APIs; webhook receiver for carrier status events.

- **Infrastructure:** AWS (ECS + RDS PostgreSQL Multi-AZ); S3 for document storage; CloudWatch dashboards; automated nightly backup with 90-day retention.

Data Migration Approach

TTS will execute a phased data migration strategy to minimize business disruption:

- **Phase 1 — Extract:** Export all master data (SKUs, suppliers, locations, customers) from the legacy system and cleanse via automated validation scripts.
- **Phase 2 — Transform:** Map legacy data to the new schema; flag exceptions for manual review; produce a reconciliation report for Meridian sign-off.
- **Phase 3 — Load:** Import cleansed master data; import last 5 years of transaction history; perform parallel-run balance reconciliation.
- **Phase 4 — Cutover:** Freeze legacy system; final delta migration; go/no-go validation checklist before live traffic switches.

4. PROJECT SCHEDULE

Phase	Deliverable	Duration	Target Date
Phase 1	Discovery, system design, data-model finalization	2 weeks	Week 2
Phase 2	Backend API & database — master data CRUD	4 weeks	Week 6
Phase 3	Inventory transactions: receive, put-away, pick, ship	4 weeks	Week 10
Phase 4	React dashboard — inventory views and reporting	3 weeks	Week 13
Phase 5	Mobile scanner app (iOS + Android)	3 weeks	Week 16
Phase 6	QuickBooks & carrier integrations	2 weeks	Week 18
Phase 7	Data migration (extract, transform, load)	2 weeks	Week 20
Phase 8	UAT, load testing, training, go-live	2 weeks	Week 22

5. PRICING

Item	Description	Hours	Rate	Cost
Discovery	Requirements, data audit, system design	80	\$125	\$10,000
Backend Dev	FastAPI, database, async workers, integrations	400	\$140	\$56,000
Frontend Dev	React dashboard, reporting, print layouts	280	\$130	\$36,400
Mobile App	React Native scanner app (iOS + Android)	160	\$130	\$20,800
Data Migration	Extract, transform, load, reconciliation	80	\$125	\$10,000

QA & Testing	Functional, integration, load, UAT support	80	\$110	\$8,800
Training	Warehouse, supervisor, and admin training	32	\$125	\$4,000
Year 1 Support	Hosting, monitoring, patches, minor fixes	N/A	Fixed	\$12,000
			TOTAL	\$158,000

Payment Schedule: 25% on contract execution (\$39,500); 25% at Phase 4 completion; 25% at Phase 7 completion; 25% at production go-live.

6. PROPOSED TEAM

Role	Name	Relevant Experience
Project Manager	Tanya Osei	10 yrs — Agile/Scrum, warehouse tech projects
Solution Architect	Dr. Priya Nair	11 yrs — AWS, distributed systems
Lead Backend Engineer	James Kowalski	10 yrs — Python, FastAPI, PostgreSQL
Frontend Engineer	Hana Sato	7 yrs — React, data-heavy enterprise UIs
Mobile Engineer	Luis Rivera	6 yrs — React Native, barcode/RFID integrations
QA Engineer	Amber Holloway	6 yrs — API testing, automated regression
Data Migration Lead	Kevin Tran	8 yrs — ETL pipelines, data quality

7. TERMS & CONDITIONS

- Proposal valid for 90 days from submission date.
- Source code and all work product become property of Meridian Distributors upon final payment.
- TTS carries \$2M general liability, \$2M professional liability, and \$1M cyber liability insurance coverage.
- Any change in scope will be documented via a Change Order and requires written authorization before additional work commences.
- Governing law: State of Tennessee.

TTS is excited about the opportunity to modernize Meridian's warehouse operations. We are confident our team has the depth and experience to deliver a reliable, scalable system that will serve Meridian for years to come. Please contact proposals@timelesstechsolutions.com with any questions.